

Hassan A. Mahmoud

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Address: G - 3932, KAUST, Thuwal, Kingdom of Saudi Arabia (KSA)

EDUCATION

King Abdullah University of Science and Technology (KAUST) – Thuwal, KSA

Doctor of Philosophy in Mechanical Engineering (Fall 2025)

Dissertation Title: “Ultra-thin, Wireless and Flexible Sensors for Smart Structures”

Cairo University, Faculty of Engineering – Cairo, Egypt

Master of Mechanical Design and Production (Fall 2021)

Thesis Topic: “Analysis of Composite Bolted Connections under Out-of-Plane Loading”

Cairo University, Faculty of Engineering – Cairo, Egypt

Bachelor of Mechanical Design and Production (Fall 2017)

Thesis Topic: “Reverse Engineering, Static and Dynamic Analysis of Turbocharger”

RESEARCH & PROFESSIONAL EXPERIENCE

 **Postdoctoral Fellow, MCEM Lab, KAUST**

November 2025 – Present

Research focuses on multifunctional printed sensing technologies and wireless monitoring solutions for composite and non-metallic infrastructure in collaboration with industry partners.

- Leading the development of fully printable wireless piezoelectric sensing platforms for structural health monitoring of mechanically demanding composite systems.
- Investigating nanoparticle-functionalized laser-induced graphene sensors for enhancing thermoelectric properties for flexible wearable cooler applications.
- Developing sensing technologies for inspection of non-metallic pipelines under the *ENERCOMP* consortium in collaboration with *Saudi Aramco*.
- Contributing to the preparation and coordination of multidisciplinary research proposals targeting smart sensing and infrastructure monitoring.

 **PhD Student, MCEM Lab, KAUST**

August 2021 – November 2025

Conducted interdisciplinary research integrating printed electronics, composite mechanics, and RF sensing to enable ultra-thin, passive, and wireless sensing platforms for structural and wearable applications.

- **Flexible E-Tattoo Wireless Sensors for Smart Structures**
 - Designed integrable e-tattoo wireless strain sensors for real-time monitoring of composite structures without compromising structural integrity.
 - Demonstrated in-situ monitoring of composite bonded joints during curing and service loading.
 - Developed low-power IoT-enabled sensing nodes for wireless data acquisition and remote structural diagnostics.
- **High Precision Wireless Strain Gauges for Structural Health Monitoring (collaboration with R&D Center, KAUST, Saudi Aramco)**
 - Developed highly sensitive, flexible, and on-demand wireless capacitive strain gauges.

- Integrated the capacitive sensors with an IoT system with wake-up radio for ultra-low power consumption.
- Performed field tests by installing sensors at Aramco facilities and collecting data wirelessly to monitor structural health and detect anomalies.
- **Fabrication Flexible Temperature Sensors using Laser-Induced Graphene (LIG)**
 - Fabricated flexible thermoelectric temperature sensors using processing-controlled Seebeck modulation in laser-induced graphene.
 - Investigated defect engineering and laser fluence effects on electromechanical properties.
 - Implemented thermopile architectures to enhance thermoelectric sensitivity.
- **Fully Printable, Flexible, and Chipless Passive Wireless Strain Sensors**
 - Designed and fabricated flexible, fully printable wireless strain sensors with ultra-high strain sensitivity for structural health monitoring of composite structures.
 - Optimized sensor linearity and sensitivity by using variable spacing and geometries in interdigitated electrode capacitive sensors.
 - Utilized e-tattoo transfer technology to apply sensors to various structures.
- **Wearable and Human-Machine Interface Sensing**
 - Developed stretchable capacitive sensors for gesture recognition in collaboration with *BrightSign Gloves (UK)*.
 - Integrated sensing systems into wearable platforms for sign language monitoring.
- **Ultrasensitive Cracked-based Wireless and Passive Strain Sensor for Structural Health Monitoring (SHM) Applications**
 - Developed ultrasensitive cracked-based passive wireless strain sensors using microfabrication technologies to detect minute strain.
 - Developed a novel design pattern to control sensor sensitivity in cracked sensors.
- **Enhanced Fracture Toughness of Composite and Hybrid Bonded Joints**
 - Performed research to enhance the fracture toughness of adhesive joints in composite materials using sacrificial cracks.
 - Developed a novel auxetic metal stiffener for metal-composite T-joints with outstanding strength and toughness performance.

Graduate Research Assistant, EXPEC ARC, Saudi Aramco

June 2025 – August 2025

Conducted collaborative research with the advanced sensing team at KAUST Upstream Research Center (KURC), EXPEC Advanced Research Center (ARC) on sensing technologies for monitoring subsurface and energy infrastructure.

- Developed and experimentally validated a wireless capacitive sensing platform for low-frequency seismic signal detection.
- Demonstrated high sensitivity within seismic frequency ranges relevant to subsurface exploration applications.

Center for Advanced Materials (CAM), The British University in Egypt (BUE), Cairo, Egypt **Graduate Research Assistant (RA)**

June 2019 – July 2021

Conducted multidisciplinary research in materials mechanics, composite structures, and renewable energy systems while contributing to student supervision and laboratory activities.

➤ **Vertical and Horizontal Axis Small Wind Turbine Designs**

- Designed and analyzed a 5 kW vertical-axis wind turbine using finite element modeling (ABAQUS), including structural assessment and composite blade layup optimization.
- Supervised composite blade manufacturing via vacuum infusion and coordinated assembly and field testing of the turbine system.
- Designed mechanical components and performed structural validation of a 10 kW horizontal-axis wind turbine, including nacelle design and composite blade development.
- Led manufacturing supervision, system integration, and experimental validation of turbine performance under operational conditions.

➤ **Optimization of Bandpass Absorption Glass Filters**

- Investigated bandpass absorption glass filters through compositional optimization using rare-earth dopants.
- Fabricated and characterized glass filters at scale to evaluate their impact on photovoltaic efficiency.
- Performed optical characterization using UV/Vis spectroscopy to determine wavelength selectivity and cutoff behavior

✚ **Mechanical Vibrations & Rotor Dynamics Lab (Prof. Aly El-Shafei's Lab), Cairo University**

Undergraduate Research Assistant

September 2016 – June 2017

Conducted experimental and computational investigations on turbomachinery components, gaining early experience in rotor dynamics, materials characterization, and multiphysics simulation.

- Performed high-resolution 3D scanning of turbocharger components and reconstructed CAD geometries using Geomagic Design X for subsequent structural and fluid analyses.
- Conducted elemental and microstructural characterization using energy-dispersive X-ray spectroscopy (EDX) to support material identification and failure assessment.
- Performed CFD and one-way fluid–structure interaction simulations of turbine and compressor rotors using ANSYS to evaluate aerodynamic performance and structural response.

GRANTS & RESEARCH FUNDING

- **“High Precision Wireless Gauges for Structural Health Monitoring”**, Awarded by ENERCOMP Consortium (KAUST & Saudi Aramco), 2024 **Total funding:** \$191,814
- **“RTF- Stretch sensors for integration into smart gloves”**, Awarded by KAUST Research Translation Grant, 2022 **Total funding:** \$751,600
- **“National Alliance for Renewable Energy Industrialization for Egypt’s Sustainable Development”**, Academy of Scientific Research and Technology (ASRT), 2018 **Total Funding:** \$ 632,892

PUBLICATIONS LIST

Journal Publications

Published

- [1] A. Abdou, **H A. Mahmoud**, M. Al Refae, Y Zhang, H. Rekik, A. Seno, G. Lubineau, “Leveraging Process-Induced Electromechanical Anisotropy of Laser-Induced Graphene for One-shot Engraving of Highly Sensitive Strain Sensors”, *ACS Applied Nano Materials* (2025), <https://doi.org/10.1021/acsanm.5c05242>
- [2] W. Badeghaish, A. Wagih, **H. A. Mahmoud**, S. Abdel-Monsef, P. Maimi, G. Lubineau, “Effect of Supercritical CO₂ Aging on the Microstructure and Fracture Properties of 3D Printed PEEK”, *Journal of Materials Research and Technology* (2025), <https://doi.org/10.1016/j.jmrt.2025.12.085>

- [3] Ahmed Wagih, F. E. Oz, R. Melentive, **Hassan A. Mahmoud**, P. Maimi, M. Abdelaziz, Gilles Lubineau, “Hydrogen Storage and Transportation Polymeric Tanks: Damage, Permeation Mitigation, and Sensing”, *International Journal of Hydrogen* (2025), <https://doi.org/10.1016/j.ijhydene.2025.151334>
- [4] **Hassan A. Mahmoud**, Abdullah AlJafari, Mohamed Bahabri, Majed A. Alrefae, Gilles Lubineau, “Processing-Controlled Seebeck Modulation in Laser-Induced Graphene for Flexible Temperature Sensors”, *ACS Applied Electronic Material* (2025), DOI: [10.1021/acsaelm.5c00562](https://doi.org/10.1021/acsaelm.5c00562)
- [5] **Hassan A. Mahmoud**, Hussein Nesser, Gilles Lubineau, “A Fully Printable Strain Sensor Enabling Highly-Sensitive Wireless Near-Field Interrogation”, *Advanced Science* (2025), DOI: [10.1002/advs.202411346](https://doi.org/10.1002/advs.202411346)
- [6] G. Lubineau, M. Alfano, R. Tao, A. Wagih, A. Yudhanto, Z. LI, K. Almuhammadi, M. Hashem, P. Hu, **H. A. Mahmoud**, F. OZ, “Harnessing extrinsic dissipation to enhance the toughness of composites and composite joints: a state-of-the-art review of recent advances”, *Advanced Materials* (2024), DOI:[10.1002/adma.202407132](https://doi.org/10.1002/adma.202407132)
- [7] **Hassan A. Mahmoud**, Hussein Nesser, A. Wagih, G. Lubineau, “Optimization Fragmentation of Metallic Film for Cracked-Based Strain Sensors”, *ACS Applied Electronic Materials* (2024), <https://doi.org/10.1021/acsaelm.4c00819>
- [8] Ahmed Wagih, **Hassan A. Mahmoud**, Gilles Lubineau, “3D printed auxetic metal stiffener for lightweight metal–composite T-joints with high strength and toughness”, *Materials & Design* (2024), <https://doi.org/10.1016/j.matdes.2024.112963>
- [9] A. Alshedayfat, A. Wagih, A. Yudhanto, **Hassan A. Mahmoud**, Gilles Lubineau, “Mode II fatigue characteristics of a composite bonded joint with microstructured adhesive bondline through tailored sacrificial cracks”, *Composite Part A: Applied Science and Manufacturing* (2024), <https://doi.org/10.1016/j.compositesa.2024.108090>
- [10] Ahmed Wagih, Harri Junaedi, **Hassan A. Mahmoud**, Gilles Lubineau, Ajay Kumar, Tamer A. Sebaey, “Enhanced damage tolerance and fracture toughness of lightweight carbon-Kevlar fiber hybrid laminate”, *Journal of Composite Materials* (2024), <https://doi.org/10.1177/00219983241235853>
- [11] Hussein Nesser, **Hassan A. Mahmoud**, G. Lubineau, “High-sensitivity RFID sensor for structural health monitoring”, *Advanced Science* (2023), <https://doi.org/10.1002/advs.202301807>
- [12] A. Wagih, **Hassan A. Mahmoud**, Ran Tao, G. Lubineau, “Towards Tough Thermoplastic Adhesive Tape by Microstructuring the Tape Using Tailored Defects”, *Polymers* (2023), <https://doi.org/10.3390/polym15020259>

Under Review/Submitted

- [1] Nidhin Divakaran, Yahya Kara, **Hassan A. Mahmoud**, Ed Hill, Gilles Lubineau, “Development of a Wearable and Sensitive PDMS-based Capacitive Strain Sensor for Sign Language Recognition”.
- [2] Mohamed S. Ibrahim, Sangho Ye, **Hassan A. Mahmoud**, Hassan Beheshti Seresht, Muhamed Y. Bayoumy, Gilles Lubineau, William R. Wanger, Youngjae Chun, “A Novel Arterial Pulsation Driven In-Situ Surface Maintenance System Using Thin Piezoelectric Layer to Enhance Hemocompatibility in Endovascular Devices”.
- [3] **Hassan A. Mahmoud**, Omar Khalifa, Yupeng Shi, Ahmed Wagih, Tareq AlNaffouri, Gilles Lubineau, “Integrable E-Tattoo Wireless Sensors for In-Situ Health Monitoring of Composite Structures”.
- [4] **Hassan A. Mahmoud**, Houssem Eddine Rekik, Gilles Lubineau, “Enhancing Linearity in Printed Capacitive Strain Sensors using Variable Spacing Interdigitated Electrodes.”
- [5] Yupeng Shi, **Hassan A. Mahmoud**, Xiaole Li, Ping Hu, Aldyandra Hami Seno, Gilles Lubineau, “Printable Multifunctional Interface Enabling Wireless Self Diagnosis: A Pathway Toward Smart Structures”

Proceedings Publications

- [1] **Hassan A. Mahmoud**, Gilles Lubineau, “Advanced E-Tattoo Wireless Strain Sensors for Smart Structural Health Monitoring of Composite Structures”, *Proceedings Volume 13435, Sensors and Smart Structures*

Technologies for Civil, Mechanical, and Aerospace Systems 2025; 134350D (2025)
<https://doi.org/10.1117/12.3052549>

- [2] **Hassan A. Mahmoud**, Gilles Lubineau, “Flexible Temperature Sensor using Laser Induced Graphene (LIG) Based on Processing-Controlled Seebeck Contrast”, *Proceedings Volume 13435, Sensors and Smart Structures Technologies for Civil, Mechanical, and Aerospace Systems* 2025; 1343517 (2025)
<https://doi.org/10.1117/12.3050755>
- [3] **Hassan A. Mahmoud**, Hussein Nesser, Gilles Lubineau, “Integrated Piezoresistivity and Frequency Modulation for High Sensitivity Printed Capacitive Strain Sensing”, *IEEE FLEPS 2024*, [DOI: 10.1109/FLEPS61194.2024.10604024](https://doi.org/10.1109/FLEPS61194.2024.10604024).
- [4] **Hassan A. Mahmoud**, Hussein Nesser, Gilles Lubineau, “Advancing Wireless Strain Sensing with Low-Cost Printable Supersensitive Radiofrequency Patches”, *e-Journal of Nondestructive Testing*,
<https://doi.org/10.58286/29725>
- [5] **Hassan A. Mahmoud**, Hussein Nesser, Ahmed Wagih, Gilles Lubineau, “High Sensitivity Wireless Strain Sensor for SHM Applications”, *Structural Health Monitoring* 2023, [DOI: 10.12783/shm2023/36847](https://doi.org/10.12783/shm2023/36847)
- [6] Gilles Lubineau, **Hassan A. Mahmoud**, Hussein Nesser, “Toward Smart Composites for Structural Health Monitoring via Highly Sensitive Capacitive Wireless Sensor”, *ECCOMAS Thematic Conference on Smart Structures and Materials (SMART 2023)*, [doi: 10.7712/150123.9766.443434](https://doi.org/10.7712/150123.9766.443434)
- [7] **Hassan A. Mahmoud**, Mostafa Shazly, Yehia Bahie El-din, Emad El-kashif, “Analysis of Composite Bolted Connection Joints under Out-of-Plane Loading”, *Proceedings of the ASME 2020 International Mechanical Engineering Congress and Exposition*. <https://doi.org/10.1115/IMECE2020-23303>.

PATENTS LIST

- [1] **H. Mahmoud**, N. Jaber, G. Lubineau, “Wireless Capacitive Sensor for Low Frequency Vibrations”, Provisional U.S. Application, Docket No. 158982-613140/SA 61314 PA (**Submitted**).
- [2] **H. Mahmoud** , O. Khalifa, M. Abdullah, A. H. Seno, N. Kouzahya, A. Asiri, A. Al-Jarro, A. Sheri, H. Saiari, T. Al-Naffouri, G. Lubineau, “Capacitive Strain Sensing IoT Device with Secure On-Demand Wireless Data Transmission”, Provisional U.S. Application, Docket No. 38136-3170001/SA 73651, Filing November 15, 2025 (**Submitted**).
- [3] **H. Mahmoud** , O. Khalifa, M. Abdullah, A. H. Seno, N. Kouzahya, A. Asiri, A. Al-Jarro, A. Sheri, H. Saiari, T. Al-Naffouri, G. Lubineau, “Data Aggregator for Reading Sensed IoT Data on Demand”, Provisional U.S. Application, Docket No. 38136-3170001/SA 73658, Filing November 15, 2025 (**Submitted**).
- [4] **H. A. Mahmoud**, Gilles Lubineau, “Controlling the Sensitivity and Linearity of Capacitive Strain Sensors Using Non-Uniform Interdigitated Electrodes”, Provisional U.S. Application No. 63/725,625. Filing November 27, 2024 (**Submitted**)
- [5] A. Wagih, **H. A. Mahmoud**, R. Tao, G. Lubineau, “Toughening Thermoplastic Adhesive Tapes using Tailored Defects”, Provisional U.S. Application No. 338-698/2022-0921-01. Filing date: August 31, 2022 (**Submitted**)

HONORS AND AWARDS

- Honorary Certificate of Appreciation (Top 20 in 2025 ComSoc Student Competition), Intelligent Structural Health Monitoring (SHM) using Semantic Edge AI and Mobile Sensing, IEEE ComSoc 2025, Taiwan **November 2025**
- Top PhD Student Ambassador at KAUST from **Qualcomm** **August 2025**
- Outstanding Performance Award, EXPEC ARC KURC, **Saudi Aramco** **August 2025**
- KAUST PSE Dean’s Research Award, **KAUST, Saudi Arabia** **May 2025**
- KAUST PSE Dean’s Research Award, **KAUST, Saudi Arabia** **June 2024**
- Best Poster Award, **LOPEC 2024, Germany** **March 2024**

TEACHING & MENTORSHIP

Teaching Assistant (TA), Mechanical Engineering, KAUST

“ME 205: Basic Laboratory Skills Course”

Fall 2023, Spring 2024, and Fall 2024

Responsibilities and Achievements:

Mentored master’s students in experimental research methods and laboratory techniques. Taught CAD and fabrication tools (3D printing, laser cutting), microscopy and image processing, oscilloscope operation, and key experiments in fluid and solid mechanics, including high-speed imaging, mechanical testing, strain gauge application, and material characterization.

Mentorship

- Mentoring **two PhD students** working on “Capacitive Sensing for In-Situ Curing and Aging Monitoring of Polymer Composites” and “Influence of Processing Parameters and Fabrication Environment on the Microstructure and Properties of Laser-Induced Graphene”. **(KAUST)**
- Mentored **two undergrad research intern students** on their projects on using LIG to develop flexible and stretchable strain and temperature sensors. **(KAUST)**

TECHNICAL EXPERTISE

Modeling, Simulation and Design

- Finite element and multiphysics simulation using ABAQUS (structural, thermal, composites, fracture and damage mechanics) and ANSYS (mechanical, CFD, thermal, Maxwell).
- Electromagnetic and RF modeling using HFSS and CST Studio; circuit-level modeling with LTSpice and COMSOL Multiphysics.
- Solid modeling and geometry processing using SolidWorks, Geomagic Design X, Avizo, MS Lattice, and Clewin
- MATLAB and Python

Experimental Fabrication and Prototyping

- Composite manufacturing techniques including hand layup, vacuum infusion, and prepreg processing.
- Micro- and nano-fabrication processes including sputtering, electroplating, wet etching, plasma RIE, photolithography, UV alignment, and laser direct writing.
- Rapid prototyping and printed electronics fabrication using polymer and SLS 3D printing, screen printing, dispenser printing (Voltera NOVA), PCB fabrication, and precision machining tools (waterjet and laser cutting).

Mechanical, Electrical, and Materials Characterization

- Mechanical testing including tensile, bending, fracture toughness, vibration testing, bulge testing, nanoindentation, and Split Hopkinson Pressure Bar experiments.
- Imaging and materials characterization using CT scanning, SEM, optical microscopy, Raman spectroscopy, and contact/non-contact profilometry.
- Electrical and RF characterization using vector network analyzers, LCR meters, impedance analyzers, probe stations, nano-voltmeters, and four-point probe measurements.

CONFERENCES & PRESENTATIONS

- [1] 19th Middle East Corrosion Conference (MECC), **Dhahran, Kingdom of Saudi Arabia, 2025** (Oral Presentation)
- [2] IEEE PIMRC, **Istanbul, Turkey, 2025** (Oral Presentation + Demo Session)
- [3] SPIE Smart Structures + NDE 2025, **Vancouver, BC, Canada, 2025** (Oral and Poster Presentations).
- [4] International Exhibition and Conference for Flexible, Organic and Printed Electronics (LOPEC), **Munich, Germany, 2025** (Poster Presentation).
- [5] IEEE International Conference on Flexible, Printable Sensors and Systems, **Tampere, Finland, 2024** (Poster Presentation).
- [6] European Workshop on Structural Health Monitoring (EWSHM), **Potsdam, Germany, 2024** (Oral Presentation).
- [7] International Exhibition and Conference for Flexible, Organic and Printed Electronics (LOPEC), **Munich, Germany, 2024** (Poster Presentation).
- [8] International Workshop on Structural Health Monitoring (IWSHM), **Stanford, US, 2023** (Oral Presentation).
- [9] 10th ECCOMAS Thematic Conference on Smart Structures and Materials (SMART 2023), **Patras, Greece, 2023** (Oral Presentation).
- [10] Energizing the Future with Composites, **KAUST, Saudi Arabia, 2023** (Poster Presentation).
- [11] European Workshop on Structural Health Monitoring (EWSHM), **Palermo, Italy, 2022** (Oral Presentation).
- [12] International Mechanical Engineering Congress and Exposition (IMECE 2020), **Virtual Conference, Online, 2020** (Oral Presentation).

COURSES & WORKSHOPS

- Structural Health Monitoring for Integrity Management of Civil Structures and Infrastructures, **Politecnico di Milano University, Italy, May 22–26, 2023**
- Basic Electromagnetic FEA, **NAFEMS, December 2022**
- Structural Health Monitoring using Statistical Pattern Recognition, **Los Alamos Dynamics, LLC, US, June 27 – July 1, 2022**
- Composites Design Workshops XXII, **Stanford University, US, January 24–28, 2022**

LANGUAGES

- **Arabic:** native language
- **English:** proficient

REFERENCES

- **Gilles Lubineau**, Professor, gilles.lubineau@kaust.edu.sa, Physical Science and Engineering Division, King Abdullah University of Science and Technology (KAUST), Thuwal, KSA
- **Tareq Al-Nafourri**, Professor, tareq.alnaffouri@kaust.edu.sa, Computer, Electrical and Mathematical Science and Engineering Division, King Abdullah University of Science and Technology (KAUST), Thuwal, KSA
- **Ahmed Wagih**, Research Scientist, ahmed.abdelhady.1@kaust.edu.sa, King Abdullah University of Science and Technology (KAUST), Thuwal, KSA
- **Nizar Jaber**, Team Lead at EXPEC ARC (KAUST Center), nizar.jaber@aramco.com, Saudi Aramco, KSA